

Worksheet: Modeling with Graphs

Group Names: _____

1. Six students are in a study group. The table says who has already worked together.

student	has worked with
Ana	Ben, Cleo, Dev
Ben	Ana, Cleo
Cleo	Ana, Ben, Dev
Dev	Ana, Cleo, Eli
Eli	Dev
Fern	—

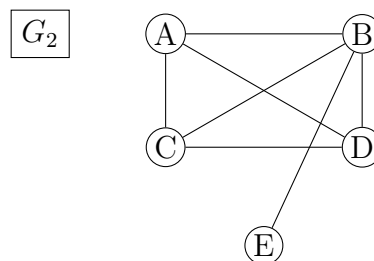
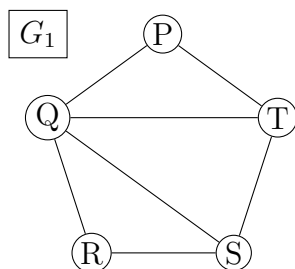
- (a) Draw a graph that represents the situation.

- (b) What are the vertices in your model? _____

- (c) What are the edges? _____

- (d) Is your graph connected? _____ What does that mean **about the study group**?

2. Below are two graphs. Two graphs are “the same” if they have the same edges (regardless of the vertex labels or how they are drawn), and they are different if there is a feature that one has that the other does not.



Same graph? _____ Why or why not? (explain)

3. We want to model the following situations.

(a) A snowplow must clear every street in a neighborhood.

vertices: _____ **edges:** _____

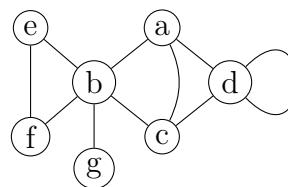
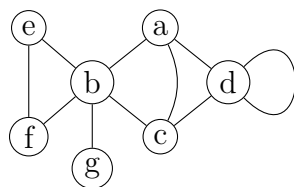
(b) A delivery driver must stop at several businesses, once each.

vertices: _____ **edges:** _____

(c) A utility wants to run fiber to every village in a region as cheaply as possible.

vertices: _____ **edges:** _____

4. **Degree.** Graph **W** is shown below, twice.



(a) Label each vertex of the **right-hand** copy with its degree. Remember a loop counts twice.

(b) Which vertex has the largest degree? _____

(c) How many edges does **W** have? _____

(d) Find a circuit and highlight/draw it on the left-hand graph. _____

(e) Find a path from e to d . Write the sequence of vertices _____

(f) Is there a vertex you could delete that would leave the graph **not** connected? _____
Why?

5. Fill in the table using the graphs on this worksheet.

graph	sum of all the degrees	number of edges
study group (#1)		
graph G_1 in #2		
graph W		

Find a pattern: the sum of all the degrees of the vertices of a graph is always _____
the number of edges.

Why should that be true?